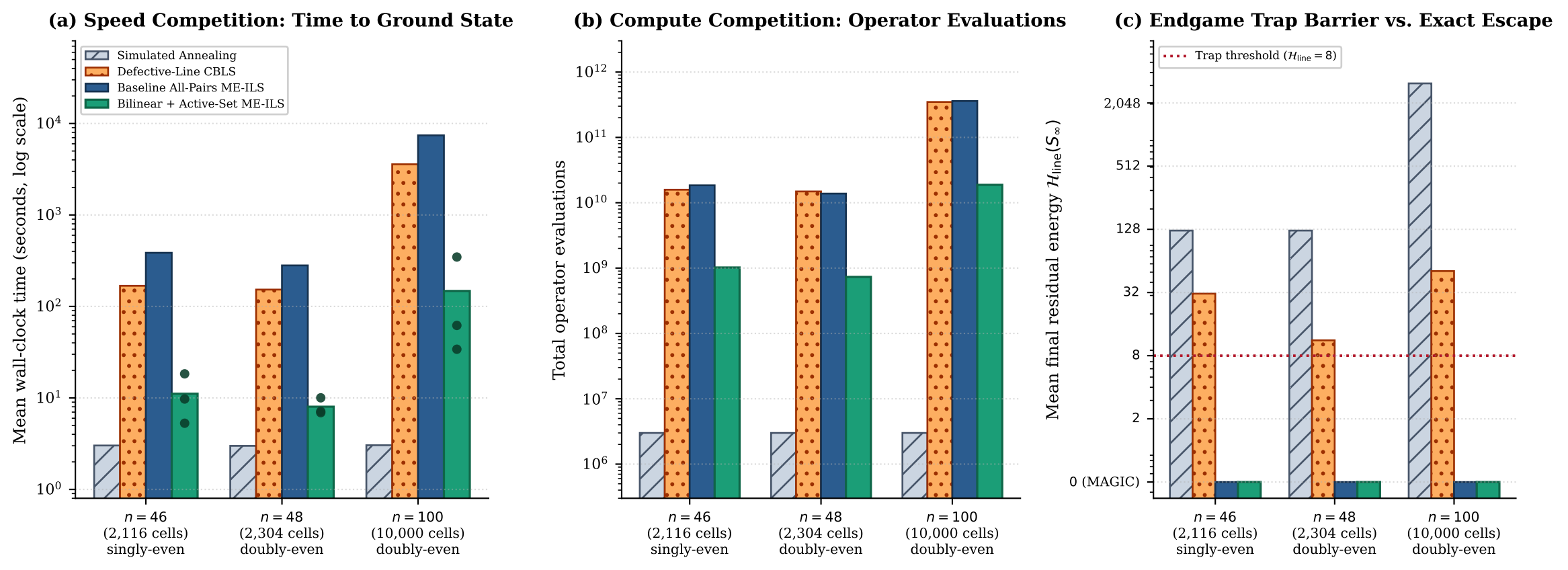

(a) Speed Competition: Time to Ground State
Simulated Annealing
Defective-Line CBLS
Baseline All-Pairs ME-ILS
Bilinear + Active-Set ME-ILS
Mean wall-clock time (seconds, log scale)
10^0
10^1
10^2
10^3
10^4
$n = 46$ (2,116 cells) singly-even
$n = 48$ (2,304 cells) doubly-even
$n = 100$ (10,000 cells) doubly-even
(b) Compute Competition: Operator Evaluations
Total operator evaluations
10^6
10^7
10^8
10^9
10^{10}
10^{11}
10^{12}
$n = 46$ (2,116 cells) singly-even
$n = 48$ (2,304 cells) doubly-even
$n = 100$ (10,000 cells) doubly-even
(c) Endgame Trap Barrier vs. Exact Escape
Trap threshold ($\mathcal{H}_{\text{line}} = 8$)
Mean final residual energy $\mathcal{H}_{\text{line}}(S_\infty)$
0 (MAGIC)
2
8
32
128
512
2,048
$n = 46$ (2,116 cells) singly-even
$n = 48$ (2,304 cells) doubly-even
$n = 100$ (10,000 cells) doubly-even